Ayush Anand

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Summary

Backend Engineer with 2+ years of experience building services in **Python** and **Java**, backed by **SQL** that stays fast at tens of millions of rows. Currently building the backend of a healthcare data platform on **Azure Kubernetes Service**: **Spring Boot** microservices, query performance tuning, and cloud-agnostic design across **Azure** and **AWS**.

Work Experience

- **Software Development Engineer (SDE) | HiLabs India Pvt. Ltd.** Sep 2025 – Present

  Healthcare data ingestion and data-quality platform for multiple US health plans, processing 1,000+ files/day across US states

  - **Clinical Data Ingestion Pipeline** | *Python, Pandas, PyArrow, MySQL*
    - Onboarded **20+** provider, lab and HIE sources via a factory-based pre-processing framework
    - Built multithreaded claims and lab **ETL** for files up to **100 GB**, with a database-driven DQ rules engine
    - Hardened ingestion against encoding, delimiter, multiline-CSV and malformed-Excel issues
    - Automated refresh of a **1.9M**-code clinical code library (ICD, CPT, LOINC, SNOMED) from UMLS releases
    - Added manual-review reason codes for AI-mapped files, surfaced in load-status reports and the UI
  - **Fallout Remediation** | *Python, MySQL, SQLAlchemy*
    - Designed and built a system that tracks whether providers fix previously failed records in later files
    - Built state-partitioned storage, a rule-based remediation engine, daily archival and remediation reports
    - Made runs idempotent and crash-resumable, consistent across reprocessing and manual rejection
    - Cut the core query from **34 hrs to 9 min (220×)** by replacing large `IN`-list scans with partition-pruned joins
  - **Member Matching** | *Python, Java, MySQL, PostgreSQL, Azure AI Search*
    - Built health-plan-specific member matching (ID-based, state-scoped, demographic) for every clinical record
    - Migrated matching from Azure AI Search to MySQL/PostgreSQL, saving ∼**$94K/year** with ∼**3**× faster lookups
    - Automated member-extract indexing, re-indexing **43M+** records with multithreaded, state-partitioned loads
  - **Reporting & Analytics** | *Python, SQL, openpyxl*
    - Built the platform's reporting layer: summary view, state breakouts, DQ and member-match fallout reports
    - Streamed multi-sheet Excel fallout reports with per-cell failure highlighting for files of **1M+** rows
    - Reconciled critical-fallout counts across dashboard, load-status and fallout reports for business users
    - Built member analytics (net-new members, coverage, penetration %) with daily and monthly rollup jobs
    - Built duplicate-file and provider layout-change detection using file- and record-level hashing
  - **Workflow Orchestration & Large-File Processing** | *Java, Spring Boot, ShedLock*
    - Built **Spring Boot** fetcher/executor **microservices** with record locking, driving each file through ∼17 stages
    - Built a self-healing framework with heartbeat-based stuck-load detection and automatic retry
    - Split files up to **100 GB** into parent/child chunk loads with stage-wise retry, load-tested at **20+** in parallel
    - Removed the Solr dependency from the Java orchestrator, deleting ∼**4K** lines of indexing code
  - **Cloud-Agnostic Platform** | *Python, Java, PostgreSQL, Liquibase, Azure, AWS*
    - Built a shared DB layer with **connection pooling**, Azure AD / AWS IAM token auth and retry with backoff
    - Abstracted cloud SDKs behind Java interfaces and Python storage handlers for Azure Blob, S3 and local
    - Migrated MySQL to **PostgreSQL on AWS RDS** using JPA/Hibernate and Liquibase, with parity testing
  - **Infrastructure, Security & Releases** | *Kubernetes, Helm, Docker, Azure DevOps*
    - Deployed on **Azure Kubernetes Service** with **Helm** and **KEDA** autoscaling, monitored via Splunk and Grafana
    - Implemented Entra ID SSO and RBAC in Spring Security, with Managed Identity for secret-free SQL/Blob access
    - Fixed VAPT/HITRUST findings, cut HIGH container CVEs by **38%**, and shipped ∼15 production releases via **CI/CD**

- **Analyst | EY Global Delivery Services** Jul 2024 – Aug 2025
  - Built the complete backend of an OPM Model application from scratch using **FastAPI**, **MySQL** and SQLAlchemy
  - Developed the backend of a Tax Cover Letter application for EY Germany using **FastAPI** and **MongoDB**
  - Built the backend of a document summarization web application using **Flask** and **MySQL**
  - Implemented discount prediction and customer segmentation models in **Python** for EY Germany

Internships

- **Software Development Intern | Noise** May 2024 – Jul 2024
  - Migrated the platform from a monolith to **microservices** and from single-tenant to **multi-tenant** architecture
  - Designed and built the Sleep 2.0 **API**, including its calculations and database migration, from scratch
- **Software Development Intern | Vayuvoy Pvt. Ltd.** May 2023 – Jul 2023
  - Built a queue management system with **NestJS** and **RabbitMQ** using priority queues over AMQP
  - Created **REST APIs** for dynamic queue creation, and consumers with acknowledgements to prevent data loss
- **Software Development Intern | Resurgent PMC** May 2022 – Aug 2022
  - Built a secure, scalable proctoring web app with **Node.js**, **Socket.io** and **Next.js** (TypeScript)
  - Added real-time monitoring with PeerJS and cheating detection with **TensorFlow.js** (phone use, multiple faces)

Projects

- **Steel Defect Detection | Bachelor Thesis Project** Mar 2024 – Apr 2024
  - Developed a steel defect detection and segmentation system using **YOLOv8** and **U-Net**
  - Improved accuracy with annotation decoding, binary masks and augmentation, and tuned BCE, Dice and IoU losses
- **Yelp Camp | Self Project** Sep 2022
  - Developed a campground review website using **Node.js**, **Express.js**, **MongoDB** and Bootstrap
  - Implemented secure user authentication and authorization, with full CRUD for campgrounds and reviews

Technical Skills

Languages: Python, Java, SQL, C++, JavaScript, TypeScript
Frameworks & Libraries: FastAPI, Flask, Spring Boot, SQLAlchemy, Pydantic, Pandas, Node.js
Databases: MySQL, PostgreSQL, MongoDB, Redis
Cloud & DevOps: Azure, AWS, Kubernetes, Docker, Helm, CI/CD, Git

Education

- **Indian Institute of Technology (IIT) Kharagpur** Dec 2020 – Apr 2024
  Bachelor of Technology in Metallurgical and Materials Engineering *GPA: 8.57/10*

Awards and Achievements

- Ranked in the top 1.6% of 2.5 Lakh+ in JEE Advanced and the top 0.6% of 10 Lakh+ in JEE Main (2020)
- Highest rating of **1798** on CodeChef and **1160** on Codeforces; solved **500+** problems across platforms
- Selected among the top **5%** of applicants for Amazon ML Summer School